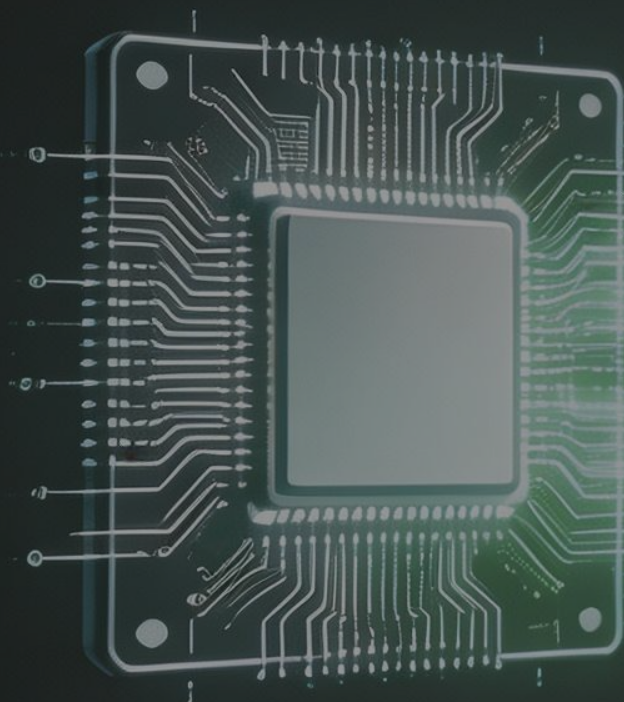




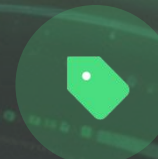
FJM30 Test

User Manual

A Fully Domestic RF Transceiver SDR
Hardware Platform



RELEASE DATE
April 24,
2026



VERSION
V 1.0.0

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Brief introduction to the FJM30 development board architecture and its core functional features.

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4.1 AD9361 Function Test

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4.2 PL DDR Test

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4.3 QSPI Firmware Test

Flash memory programming verification and secure boot sequence functional testing.

Board Overview



Hardware Specs

Product Description

A fully domestic RF transceiver SDR platform. Internally integrates high-performance ARM, Xilinx XC7Z020 FPGA, and AD9361 frequency agile chip modules.



Target Scenarios

Key Applications

- ✓ 3G/4G/5G macrocellular Time Division Duplex (TDD) base stations.
- ✓ High-precision wireless communication test equipment.
- ✓ Flexible software-defined radio (SDR) research applications.



Core Highlights

Core Advantages

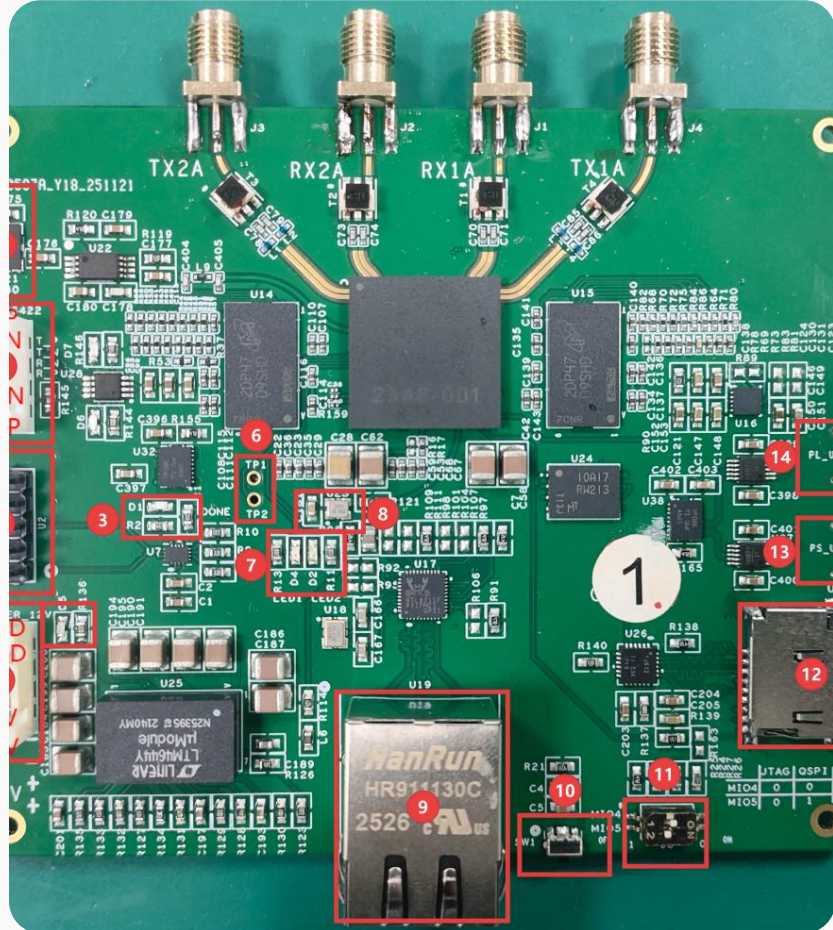
High Programmability

Offers extreme flexibility for custom transceiver algorithm development.

Broadband Capability

Covers a wide range of frequency bands, ideal for multi-standard applications.

2. Board Hardware Structure



01

Power Supply Interface
12V DC input & Power LED

02

JTAG Debug Interface
Standard 10-pin for debugging

03

Program Done Indicator
LED for FPGA download status

04

PL-side RS422 Port
High-speed serial communication

05

Reference Input (40MHz)
On-board external clock source

06

Test Points (2pcs)
For signal measurement & debug

07

Custom LED Indicators
2 user-configurable status LEDs

08

PS-side Clock Source
Stable 33.3333MHz system clock

09

10/100M Ethernet Port
RJ45 connector for network access

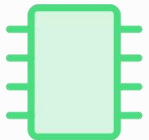
10

System Reset Button
Hard reset for FPGA/SoC system

3. Patch Installation

PURPOSE

To compile and modify programs using Xilinx Vivado for the 58th Research Institute's FPGA chip, a specific patch is required for compatibility.



INSTALLATION STEPS



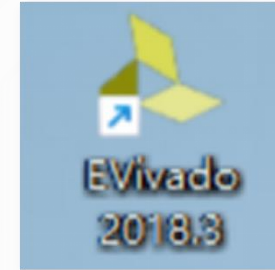
01. Locate Folder

Find the designated "Software Patch" directory.

02. Execute Installer

Follow the detailed instructions inside the folder.

VERIFICATION



Desktop Shortcut

A new icon "Evivado 2018.3" will appear.

Mandatory Usage

Use only this version for all subsequent tasks.

4.1 AD9361 Function Test (TX)



OBJECTIVE

Verify the transmit (TX) signal generation functionality of the AD9361 RF transceiver chip.



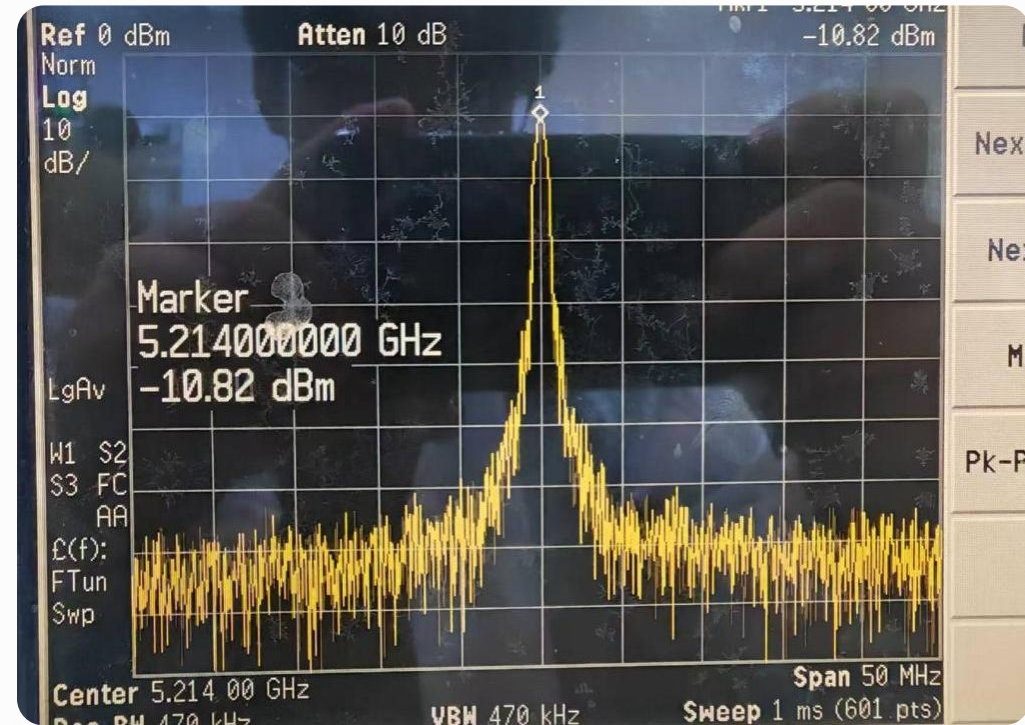
EXPERIMENT SETUP

1. Navigate to `test\_9361` & open in Vivado 2018.3.
2. Program the FPGA with the pre-built bitstream.



TRANSMIT (TX) VERIFICATION

Connect TX1A/TX2A to a spectrum analyzer. Observe the output signal at the default frequency of **5.214 GHz**.



Measurement Result: **5.214 GHz Peak Detected**

4.1 AD9361 Function Test (RX)

RECEIVE TEST WORKFLOW



Hardware Setup

Connect a signal generator output to the board's RX1A and RX2A SMA ports for differential input.



Signal Generator Configuration

Freq: 5.214 GHz (Sync w/ TX LO)
Amp: -30.00 dBm (Input Level)



Receiver Calibration Parameters

Bandwidth: 500 KHz | Sampling Rate: 40 MHz

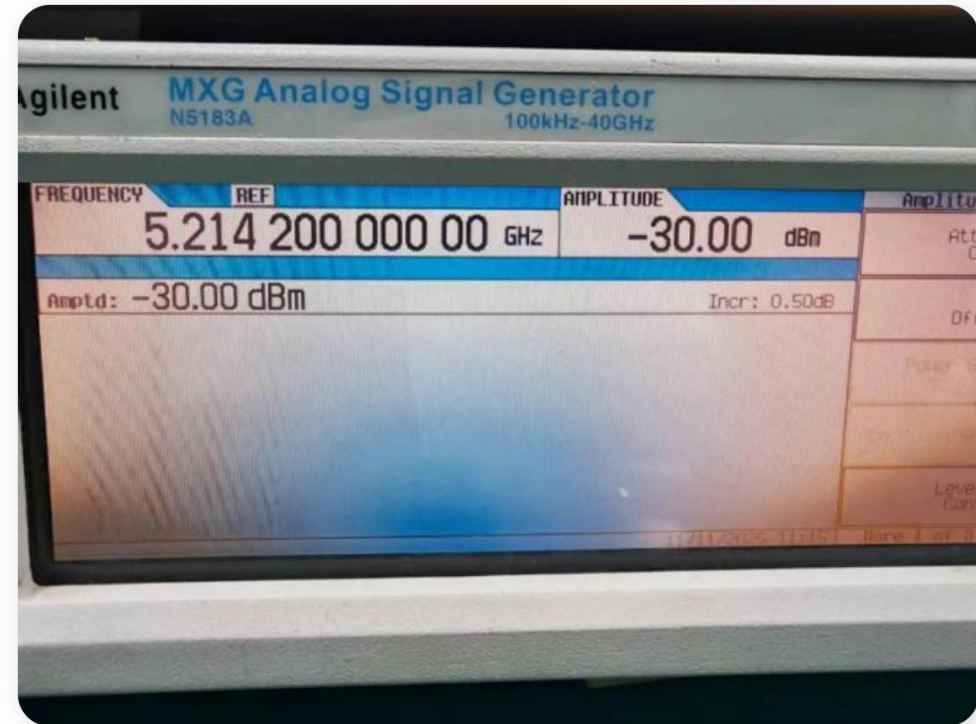


Figure 4.1: Signal Generator UI showing the configured frequency and amplitude.

4.1 AD9361 Function Test (Waveform Verification)

Verification via ILA (Integrated Logic Analyzer)



Waveform Observation

Utilize the ILA tool within Vivado to visually inspect and monitor the received real-time waveform data.



Key Test Condition

Input signal is configured with a precise 200 KHz frequency offset relative to the Local Oscillator (LO).



Validation of RX Path

At 40 MHz sampling rate, 200 samples per period are captured, confirming the RX path is operating as expected.



Figure 1: ILA Waveform Capture in Vivado

4.1 AD9361 Function Test (Frequency Change)

How to Change LO Frequency



TDD Mode Requirement

The chip only supports TDD mode, so the TX and RX frequencies must be configured to the same value.



VIO Interface Setup

In the VIO panel, set identical numerical values (in Hz) for the `txio` and `rxio` input registers.



Apply the New Configuration

1. Pull `rxio\_en` & `txio\_en` LOW. 2. Pull them HIGH again to trigger the frequency reload.

	Value	Acti...	Directi...	VIO
et	1		Output	hw_vio_1
	1		Output	hw_vio_1
	[U] 20		Output	hw_vio_1
a[31:0]	[H] 0002_36F8		Input	hw_vio_1
	[H] 02_B000		Output	hw_vio_1
	[B] 0		Output	hw_vio_1
	[B] 1		Output	hw_vio_1
	[B] 1		Output	hw_vio_1
	[U] 4200000000		Output	hw_vio_1
	[U] 4200000000		Output	hw_vio_1
	1		Output	hw_vio_1
	1		Output	hw_vio_1
d[15:0]	[H] 0000		Output	hw_vio_1
	[B] 1		Output	hw_vio_1
	[B] 0		Output	hw_vio_1
)]	[H] 0000_0000		Output	hw_vio_1
	[B] 0		Output	hw_vio_1
	[B] 0		Output	hw_vio_1
d_en	[B] 0		Output	hw_vio_1
0]	[H] 1		Output	hw_vio_1
	[U] 100		Output	hw_vio_1
	[U] 100		Output	hw_vio_1
	1		Output	hw_vio_1

4.1 AD9361 Function Test (Baseband Signal Modification)

```

JM30_test_v2/test_9361/sources_9361n/src/top.v

0m),

rx_channel_sel),
tx_channel_sel))

(sys_clk),

(A_sample_clk),
(12'h'ff),
(12'h'ff),
(12'h'ff),
(12'h'ff),
(A_sample_clk),
(A_rx_data_i1),
(A_rx_data_q1),
(A_rx_data_i2),
(A_rx_data_q2),
(A_data_clk),

(A_rx_clk_in_p ),
(A_rx_clk_in_n ),

(A_rx_frame_in_p ),
(A_rx_frame_in_n ),
(A_rx_data_in_p ),
(A_rx_data_in_n ),
(A_tx_clk_out_p ),
(A_tx_clk_out_n ),
(A_tx_frame_out_p),
(A_tx_frame_out_n ),

```

基带信号

AD数据时钟

AD数据



Code Replacement

The baseband signal source can be replaced directly in the Verilog source code to change the transmission data.



Locate `tx\_data` Definition

Find the exact section in the code where the `tx\_data` array is declared, which holds the baseband signal values.



Inject Custom Signal Data

Replace the default hard-coded numerical values with your own custom IQ data to test different transmission scenarios.

4.1 AD9361 Function Test (Other VIO Controls)

Additional VIO Functionality

For comprehensive details on other functions controllable via the VIO interface, please refer to the detailed comments provided within the `GMD9361\_top` module source code.

C:/cd/PGA/FPGA/JM30F/JM30F_testTest_9361/insrc/GMD9361_top.v

```
55 input [11:0] data_pi,
56 `endif
57 input tx_sample_clk,
58 input [11:0] tx_data_i1,
59 input [11:0] tx_data_i2,
60 input [11:0] tx_data_i3,
61 input [11:0] tx_data_i4,
62 output rx_sample_clk,
63 output [11:0] rx_data_o1,
64 output [11:0] rx_data_o2,
65 output [11:0] rx_data_o3,
66 output [11:0] rx_data_o4,
67 output data_11k,
68 `ifdef DC706_33M0_TEST //以下为用户传入的芯片控制信号，不使用的控制接口可以在编译时给值0。
69 input gm40361_reset, //复位信号必须提供，低电平复位，至少保持6个时钟，再拉高
70 input [8:0] tx_tx_freq, //发射本振设置频率，设置范围30M-6000M
71 input [8:0] tx2_attn, //tx2衰减设置44(可设置范围0-359, 代表步进0.25dB+设置值的衰减)，不设置时零0;
72 input [8:0] tx2_attn, //tx2衰减设置44(可设置范围0-359, 代表步进0.25dB+设置值的衰减)，不设置时零0;
73 input tx1_iq_cal_in, //tx1校准信号，上升沿触发校准，即先给0再给1
74 input tx2_iq_cal_in, //tx2校准信号，上升沿触发校准，即先给0再给1
75 input rx_tx_freq, //接收本振设置频率，设置范围30M-6000M
76 input [7:0] rx1_gain, //MISO模式Tx1初始化和默认增益值，可设置范围0-73, 步进1dB
77 input [7:0] rx2_gain, //MISO模式Tx2初始化和默认增益值，可设置范围0-73, 步进1dB
78 input rx1_iq_cal_in, //rx1校准信号，上升沿触发校准，即先给0再给1
79 input [1:0] gain_mode_in, //增益模式设置，范围0-0 (其中0=MISO, 1=Fast_AGC, 2=Slow_AGC)
80 input [15:0] gain_threshold_in, //AGC阈值设置，范围如下:
81 //当rx_gain_mode=0时使用慢增益模式 (Slow_AGC)，第8位15:8表示阈值上拉，第8位表示阈值下拉，单位-dBFS;
82 //当rx_gain_mode=1时使用快速增益模式 (Fast_AGC)，第8位15:8表示阈值，单位-dBFS;
83 input [31:0] rx_bb_in, //接收带外设置频率，范围200K到900M.
84 `endif
```

Important Note: Enable Signal Toggle

Any parameter changes made through the VIO interface will not take effect immediately. The corresponding `en` (enable) signal must be explicitly toggled (pulled low, then high) to trigger the update logic.

```
160 assign tx2_attn_dc = tx2_attn;
161 assign gain_mode = tx_gain_mode;
162 assign gain_threshold = tx_gain_threshold;
163 assign tx1_iq_cal = tx1_iq_cal;
164 assign tx2_iq_cal = tx2_iq_cal;
165 assign rx_bb_bw = rx_bb_bw;
166 assign rx_iq_cal = rx_iq_cal;
167 `else
168 assign sys_rst = tx_reset_en? tx_gm40361_reset: gm40361_reset;
169 assign pin_rst_n = tx_reset_en? tx_gm40361_reset: gm40361_reset;
170 assign rx_lo = tx_rx_lo_en? tx_rx_lo_freq:
171 // rx_lo = 64'd161_000_000;
172 assign rx1_gain_db = tx_rx1_gain_en? tx_rx1_gain: rx1_gain;
173 assign rx2_gain_db = tx_rx2_gain_en? tx_rx2_gain: rx2_gain;
174 assign tx_lo = tx_tx_lo_en? tx_tx_lo_freq:
175 // tx_lo = 64'd161_000_000;
176 assign tx1_attn_dc = tx_tx1_attn_en? tx_tx1_attn: tx1_attn;
177 assign tx2_attn_dc = tx_tx2_attn_en? tx_tx2_attn: tx2_attn;
178 assign gain_mode = tx_gain_mode_en? tx_gain_mode: gain_mode;
179 assign gain_threshold = tx_gain_threshold_en? tx_gain_threshold: gain_threshold;
180 assign tx1_iq_cal = tx_tx1_iq_cal_en? tx_tx1_iq_cal: tx1_iq_cal;
181 assign tx2_iq_cal = tx_tx2_iq_cal_en? tx_tx2_iq_cal: tx2_iq_cal;
182 assign rx_bb_bw = tx_rx_bb_bw_en? tx_rx_bb_bw: rx_bb_bw;
183 assign rx_iq_cal = tx_rx_iq_cal_en? tx_rx_iq_cal: rx_iq_cal;
184 `endif
185 `else
186 assign sys_rst = gm40361_reset;
187 assign pin_rst_n = gm40361_reset;
```

此处改动生效的前提是en拉高

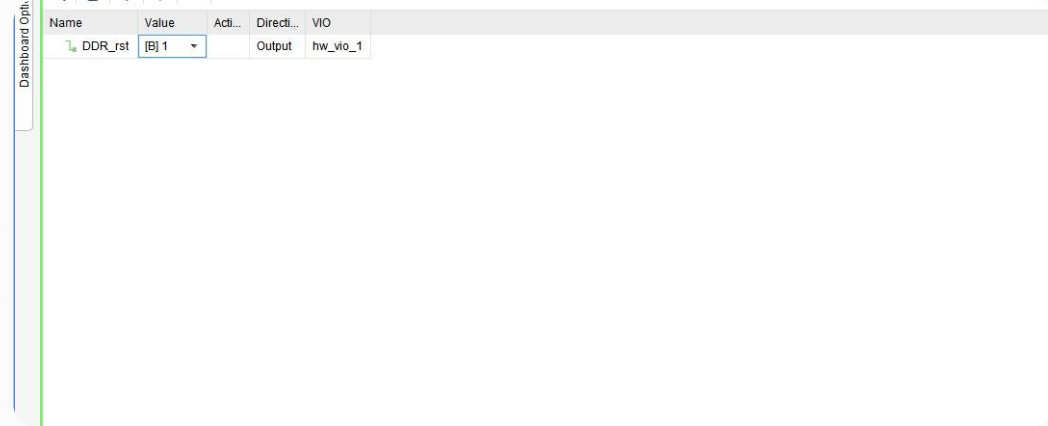
4.2 PL DDR Test

OBJECTIVE

To verify the functionality of the DDR memory connected to the FPGA's PL (Programmable Logic) side.

TEST STEPS

- 1 Navigate to the `test\_PL\_DDR` folder and open the project.
- 2 Program the FPGA device.
- 3 Toggle `DDR\_rst` in VIO (low → high) to initiate initialization.
- 4 Check ILA waveform for `DDR\_init\_complete` (High).



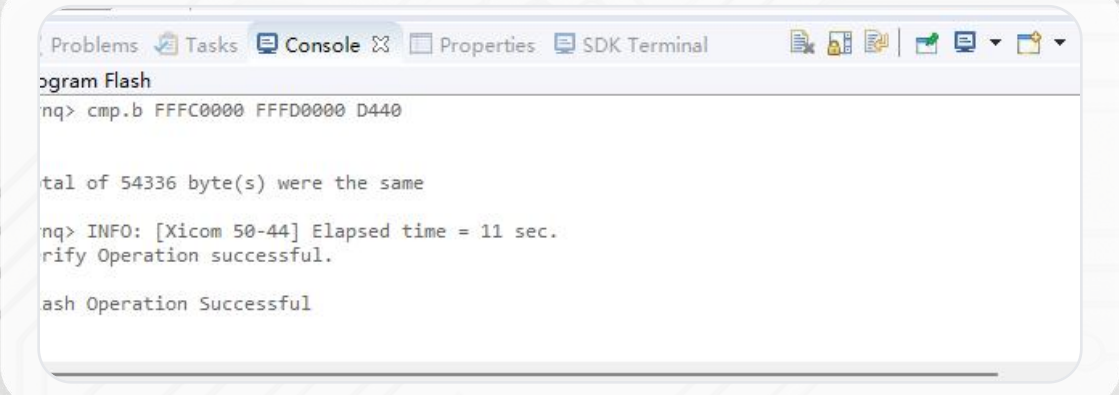
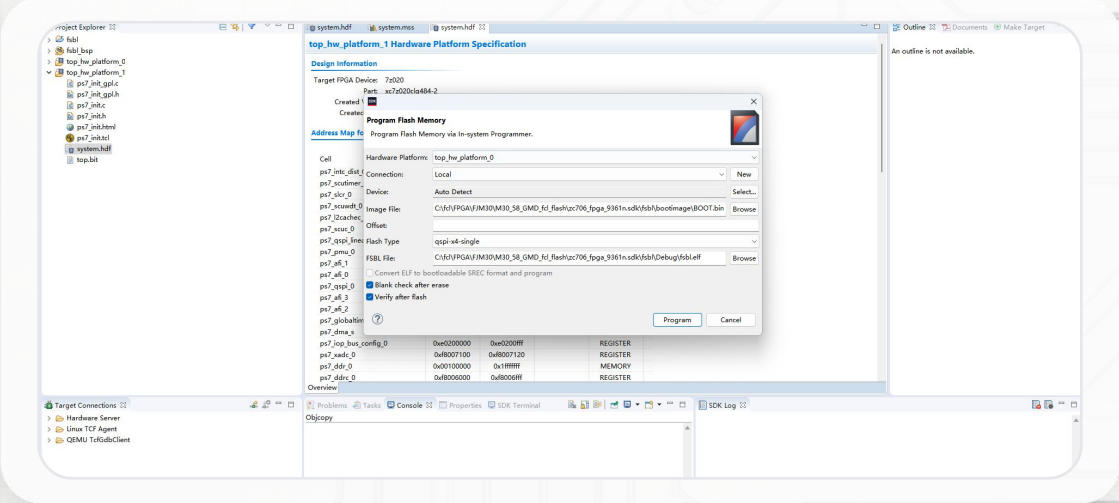
4.3 QSPI Firmware Test (Programming)

OBJECTIVE

To test the process of programming the FPGA configuration file into the QSPI flash memory for standalone boot.

PROGRAMMING STEPS

- 1 Navigate to the `test\_QSPI` folder and open the project.
- 2 Launch the Xilinx SDK software.
- 3 Click on the `Program Flash` option in the menu.
- 4 Browse and select the appropriate `.bin` file in the dialog box.



4.3 QSPI Firmware Test (Verification)

Verification Steps

01

Power Down the Board

After the flash programming operation completes successfully, turn off the power supply to the board.

02

Set QSPI Boot Mode

Adjust the DIP switch (Item ⑪) on the board to the correct position that enables booting from the QSPI flash memory.

03

Power the Board Back On

Reconnect the power supply to the board. The system will now attempt to initialize using the QSPI flash.

04

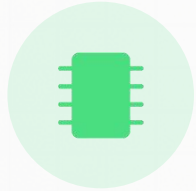
Check the DONE LED

Observe the FPGA Program Download Completion Indicator (Item ③). If it lights up, the QSPI boot is successful.



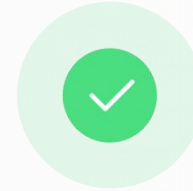
Note: This process is confirmed visually by observing the DIP switch position and the status of the DONE indicator LED.

Summary



Versatile SDR Platform

The FJM30 is a fully domestic SDR solution built around the Zynq-7020 SoC and AD9361 transceiver, offering a flexible foundation for wireless research.



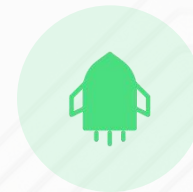
Core Functionality Validated

Rigorous testing confirms stable RF Tx/Rx operation, reliable PL DDR memory access, and successful QSPI flash programming for standalone booting.



Verified Development Toolchain

All system validation was conducted using Xilinx Vivado 2018.3 with a critical platform-specific patch, ensuring compatibility and reproducibility.



Ready for Deployment

The FJM30 platform is fully tested and ready for integration into advanced wireless communication systems, supporting rapid prototyping and product development.

Thank you

